KAPASA MAKASA UNIVERSITY

SCHOOL OF NATURAL RESOURCES

DEPARTMENT OF AGRICULTURE AND AQUATIC SCIENCES

2024/25 SESSIONAL EXAMINATIONS

SA340

SOIL AND WATER MANAGEMENT

INSTRUCTION: 1. There are seven questions in total

2.Answer a compulsory question in **SECTION A**

3. Answer any four (4) questions out of the six in **SECTION B**

4. There are 100 marks allocated to this paper

5. Where possible illustrate your answers

7. When answering follow further instructions in the answer boo

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SECTION A: *Compulsory Question-40 Marks*

Question No.1

Mr. John Banes Chomba recently retired and upon getting his pension he acquired a 200-hectare farm in the interior of the Manshya Farm block of Chinsali District. He intends to go into production of fruits and vegetables, and the average rainfall around his farm is below normal. He further engages you as a fresh graduate of soil and water management from Kapasa Makasa University as a consultant over issues of Irrigation, Soil texture and also the status of the soils in terms of Infiltration rates, at his farm.

Earlier on before engaging you, a college student had conducted an experiment where a field sample of an unconfined aquifer was packed in a cylinder (with length 60cm and diameter 8cm), the field sample was tested for 4 minutes under constant head difference of 15. 3cm. Then 43.7cm^3 of water was collected at the outlet and also determined the pore space as 55%. So based on your capabilities as a fresh graduate, clarify on the following to Mr. John Banes Chomba as your client;

(a) Explain to your client why Irrigation is important in fruit and vegetable production. ***(4 marks)***

(b) What kind of Irrigation System can you recommend to your client and why? ***(4 marks)***

(c) With the aid of a well labelled illustration briefly explain the irrigation design system you have recommended and highlight the functions of its major components. ***(10 marks)***

(d) What are the main advantages and disadvantages of the Irrigation System recommended in (b)? ***(5 marks)***

(e) Briefly explain to your client the ***three*** major methods of determination of Soil texture on his farm? ***(6 marks).***

(f) Based on the data generated by the earlier college student,

(i) Determine the hydraulic conductivity of the aquifer sample? ***(6 marks).***

(ii) Compute the bulk density and comment on the implication of its value as regards to crop production? ***(5 marks)***

SECTION B: ***Answer any 4 Questions*** **(out of the 6).** ***Each question carries 15 marks***

Question No. 2

Define the following terms, (from i to ii)

(i) Cation Exchange Capacity (CEC) ***(2 marks)***

(ii) Isormophous Substitution ***(2 marks)***

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(iii) A soil from KMU contains 38% clay and has a CEC of 1.67 cmol (+)/kg soil. Calculate the CEC of the clay fraction assuming the measured CEC is due to clay only and state the probable clay type *(6marks)*

(iv) A soil has a CEC of 15 cmol (+)/kg. Calculate the amount of exchangeable calcium (Ca^{2+}) it retains in kg/ha if the soil has a depth of 200mm and a bulk density of 1300 kg/m^3? *(5 marks)*

$$\frac{0.082 \times 1.67}{0.38}$$

Question No.3

(i) Soil color is one property that can be used to describe horizonation and soil morphology. Discuss. *(10 marks)*

(ii) State and define the 3 properties used to describe soil color using the Munsell system. *(3marks)*

(iii) Give an example of a Munsell notation clearly indicating the properties stated in (i) above. *(2marks)*

Question No.4

Describe the roles of the ***Dokuchaeiv's*** soil forming factors in the formation of soils and their interrelationships. *(15 marks)*

Question No.5

(i) Compare and contrast the ***Matric Potential*** (ψ_m) Vs ***Osmotic Potential*** (ψ_o) **(5 marks)**

*(ii)*A meristematic plant tissue with a volume of $0.0008mm^3$ was placed in an open beaker of 0.4M sucrose at 25°C. What is the water potential of the solution in the beaker? *(5 marks)*

0.082

*(iii)*If a plant tissue above has a water potential of -0.8 Mpa. Which way will the water move? How will this affect the cell? *(5 marks)*

QUESTION No.6

(i) Briefly explain the mechanism of the Mycorrhiza Relationship? *(5 marks)*

(ii) Discuss the functions of soil micro-organisms in relation to soil health and fertility. *(10 marks)*

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QUESTION No.7

Define the following terms, (from i to ii)

(i) Gravimetric Moisture Content (O_g) ***(2 marks)***

(ii) Volumetric Moisture Content (O_v) ***(2 marks)***

(iii) Calculate the porosity of a soil sample that has a bulk density of 1.35 g/cm^3. Assume the particle density is 2.65 g/cm^3. ***(6marks).***

(iv) Calculate the bulk density of a soil sample when ¾ of the pores are at saturation. ***(5marks)***

-END OF EXAMINATION-

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KAPASA MAKASA UNIVERSITY

COURSE: SOIL AND WATER MANAGEMENT –SA340 TEST II

DURATION: 1.5 HRS

INSTRUCTIONS: Answer all questions and be precise.

Question No.1

Define the following terms;

(i) Nitrification *(2 marks)*
(ii) Flocculation *(2 marks)*
(iii) Ion Adsorption *(2 marks)*
(iv) Deprotonation *(2 marks)*
(v) P^{H} Dependent Charge *(2 marks*

Question No.2.

A soil has a CEC of 15 cmol (+)/kg. Calculate the amount of exchangeable calcium (Ca2+) it retain in kg/ha if the soil has a depth of 20 cm and a bulk density of 1300 kg/m^3? *(15 marks)*

Question No. 3

Define the following terms, (from i to iii)

(i) Cation Exchange Capacity (CEC) *(2 marks)*
(ii) Isormophous Substitution *(2 marks)*
(iii) E-Horizon *(2 marks)*
(iv) A soil from KMU contains 37 % clay and has a CEC of 1.72 cmol (+)/kg soil. Calculate the CEC of the clay fraction assuming the measured CEC is due to clay only. *(10marks)*

Question No.4

The prudent use of soils depends on recognition of soil properties and predictions of soil behavior under various conditions.

a. Using the Munsell system, color is described in terms of three (3) properties.

(i) State and define the 3 properties used to describe soil color using the Munsell system. *(6marks)*

(ii) Give an example of a Munsell notation clearly indicating the properties stated in (i) above. *(4marks)*

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Question No.5

Compare and contrast the architecture and description of the Soil horizons in ecological region III and region I of Zambia, support your answer with an illustration. ***(10 marks)***

Question No.6

The micro-nutrients are required in smaller quantities for plant growth. But they are very vital in ensuring a normal and healthy plant growth. State at least ***two*** major functions of the following micronutrients, ***(5marks)***

(i) Nickel
(ii) Boron
(iii) Zinc
(iv) Chlorine
(v) Manganese

QUESTION No.7

(i) Briefly explain the mechanism of the Mycorrhiza Relationship? ***(5 marks)***

(ii) Discuss the functions of soil micro-organisms in relation to soil health and fertility. ***(10marks)***

-END OF TEST-

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